

On a Packing and Covering Problem

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Let positive integers $r < k < N$ and a family $\mathcal{F}$ of k -element subsets of $\{1, 2, \dots, N\}$ be given. We say that $\mathcal{F}$ is r -dense if any r -element subset of $\{1, 2, \dots, N\}$ is contained in at least one member of $\mathcal{F}$. On the other hand we say that $\mathcal{F}'$ is r -sparse if any two members of $\mathcal{F}'$ intersect in less than r -elements—i.e. every r -element subset of $\{1, 2, \dots, N\}$ is contained in at most one member of $\mathcal{F}'$. It is well known (see [4]) that

$$|\mathcal{F}| \geq \frac{\binom{N}{r}}{\binom{k}{r}} \geq |\mathcal{F}'| \quad (1)$$

for any r -dense family $\mathcal{F}$ and r -sparse family $\mathcal{F}'$, $\mathcal{F}, \mathcal{F}' \subset [\{1, 2, \dots, N\}]^k$. In [2] Erdős and Spencer denote by $M(N, k, r)$ the minimal number of elements of r -dense family $\mathcal{F} \subset [\{1, 2, \dots, N\}]^k$ and by $m(N, k, r)$ the maximal number of elements of r -sparse family $\mathcal{F}' \subset [\{1, 2, \dots, N\}]^k$. From (1) we get

$$M(N, k, r) \geq \frac{\binom{N}{r}}{\binom{k}{r}} \geq m(N, k, r).$$

It was shown by P. Erdős and J. Spencer [2] that

$$M(N, k, r) \leq \left[\frac{\binom{N}{r}}{\binom{k}{r}} \right] [1 + \log \binom{k}{r}].$$

In 1963 P. Erdős and J. Hanani [1] conjectured that

$$\lim_{N \rightarrow \infty} M(N, k, r) \left(\frac{k}{r} \right) \left(\frac{N}{r} \right)^{-1} = \lim_{N \rightarrow \infty} m(N, k, r) \left(\frac{k}{r} \right) \left(\frac{N}{r} \right)^{-1} = 1, \quad (2)$$

for every fixed r and k , $r < k$. They proved (2) for $r = 2$ and all k and for $r = 3$, $k = p$ or $p + 1$ where p is power of a prime.

The objective of this paper is to prove (2) (cf. [5], where a few related remarks are mentioned).

PRELIMINARIES

We find it convenient to work with the following structures. Let J be a k -element set (k -set) of positive integers. A k -partite r -graph is a pair

$$G = ((V_j)_{j \in J}, E)$$

such that $|e \cap V_j| \leq 1$ for every $j \in J$ and $e \in E$, moreover $|e| = r$ and $e \subset \bigcup_{j \in J} V_j$ for every $e \in E$.

If $G = ((V_j)_{j \in J}, E)$ then we set $V(G) = \bigcup_{j \in J} V_j$, $E(G) = E$. For $I \in [J]'$ ($[J]'$ denotes the set of all r -element subsets of J) $\rho_I = \rho_I(G)$ denotes the cardinality of $E_I(G)$, where

$$E_I(G) = \{e \in E; e \cap V_i \neq \emptyset \text{ for every } i \in I\}.$$

We say that subset $L \subset \bigcup_{j \in J} V_j$, $|L| \geq r$ is *complete set* if $[L]^r \subset E$. Let L be a complete set of cardinality k , we say that the set $K = [L]^r$ forms a *k-gon*. Let L be a complete set in G , then $\sigma^L(G)$ will denote the number of complete k -sets containing L as a subset.

Let $A_1, A_2, \dots, A_p$, $A = \bigcup_{i=1}^p A_i$ be pairwise disjoint sets. By the symbol $[\{A_i\}_{i=1}^p]^r$ we shall denote the system of all r -subsets of A , intersecting each A_i in at most one element. We shall often use the symbol $o(1)$ to denote a quantity (depending on the natural number n) the value of which tends to zero as n (here always n) tends to infinity. For a function $f(n)$ we put $o(f(n)) = o(1)f(n)$. For two functions $f(n), g(n)$ we will write $f(n) \sim g(n)$ if $f(n) = (1 + o(1))g(n)$. We also write $f(n) \sim_{\delta} g(n)$ if $|f(n) - g(n)| < \delta f(n)$. Finally we will state here the following auxiliary.

CLAIM. *For every pair of positive integers n, m , $n > m$ and positive reals p, q , $p + q = 1$ such that*

$$(2p - 1)n < m < 2pn$$

we have

$$\binom{n}{m} p^m q^{n-m} < \exp\left(-\frac{1}{3} \frac{(m - np)^2}{npq}\right). \quad (3)$$

We omit the standard proof based on Stirling formula. (For the proof of very similar statements see e.g. [3]).

RESULTS

The objective of this paper is to prove the following theorem.

THEOREM. *For every pair of positive integers r and k , $r < k$,*

$$\lim_{N \rightarrow \infty} M(N, k, r) \binom{k}{r} \binom{N}{r}^{-1} = \lim_{N \rightarrow \infty} m(N, k, r) \binom{k}{r} \binom{N}{r}^{-1} = 1$$

holds.

First we show the easy fact that

$$\lim_{N \rightarrow \infty} M(N, k, r) \binom{k}{r} \binom{N}{r}^{-1} = 1$$

implies that

$$\lim_{N \rightarrow \infty} m(N, k, r) \binom{k}{r} \binom{N}{r}^{-1} = 1$$

and thus that it will be sufficient to show the first statement only:

Let

$$\mathcal{F} \subset [\{1, 2, \dots, N\}]^k, |\mathcal{F}| \leq \binom{N}{r} \binom{k}{r}^{-1} (1 + \delta)$$

be an r -dense family. For every $e \in [\{1, 2, \dots, N\}]^r$ let $\nu(e)$ be the number of elements of $\mathcal{F}$ containing e . We have

$$\sum \{\nu(e); e \in [\{1, 2, \dots, N\}]^r\} \leq (1 + \delta) \binom{N}{r}$$

and thus

$$\sum \{\nu(e); \nu(e) \geq 2\} \leq 2\delta \binom{N}{r}.$$

After deleting (from $\mathcal{F}$) all elements of F containing r -sets e with $\nu(e) \geq 2$ we get at least

$$\binom{N}{r} \binom{k}{r}^{-1} (1 + \delta) - 2\delta \binom{N}{r} = \binom{N}{r} \binom{k}{r}^{-1} \left(1 + \delta - 2\delta \binom{k}{r}\right)$$

k -sets, forming an r -sparse family. The last quantity tends to

$$\binom{N}{r} \binom{k}{r}^{-1} \text{ as } \delta \rightarrow 0;$$

this proves our claim.

The proof of the theorem is divided into three lemmas.

LEMMA 1. Let $G = ((V_i)_{i=1}^k, E)$, $|V_1| = |V_2| = \dots = |V_k| = n$ be a k -partite r -graph, ρ and $\sigma_l (l = r, \dots, k-1)$ positive reals smaller than one such that

(α) $\sigma^L(G) \sim \sigma_l n^{k-l}$ for any L , complete (in G) $k > |L| = l \geq r$.

(β) $\rho_I(G) \sim \rho n^r$ for any $I \in [\{1, 2, \dots, k\}]^r$.

Then for every $\varepsilon > 0$ one can select a system $\mathcal{K}$ of k -gons from G such that if we put

$$\begin{aligned} G^* &= ((V_i)_{i=1}^k; E - \{e; \exists K \in \mathcal{K}, e \in K\}), \\ \rho_I^* &= \rho_I(G^*), \\ \sigma^{L*} &= \sigma^L(G^*), \end{aligned}$$

the following hold:

- (a) $\rho_I^* \sim (\rho \exp(-\sigma_r \varepsilon)) n^r$ for any $I \in [\{1, 2, \dots, k\}]^r$.
- (b) $\sigma^{L*} \sim \left[\sigma_l \exp\left(-\sigma_\mu \varepsilon \left(\binom{k}{r} - \binom{l}{r}\right)\right) \right] n^{k-l}$ for any L complete (in G^*), $k > |L| = l \geq r$.
- (c) $|\{\{K^1, K^2\}, K^1, K^2 \in \mathcal{K}, K^1 \cap K^2 \neq \emptyset\}| \leq 2\sigma_r \varepsilon |\bigcup_{K \in \mathcal{K}} K|$.

REMARK. Roughly speaking this Lemma asserts that for a given $\varepsilon > 0$ there is $n > n_0(\varepsilon)$ such that if $\sigma^L(G)$, $\rho_I(G)$ are close to values described in (α)(β) one can select a system $\mathcal{K}$ with ρ_I^* , σ^{L*} close to values described in (a) and (b) and moreover such that (c) holds.

PROOF OF LEMMA 1. Let $G = ((V_i)_{i=1}^k, E)$ be a given k -partite r -graph with the properties (α) and (β) of Lemma 1. Suppose without loss of generality that $n = |V_1| = |V_2| = \dots = |V_k|$ is large positive integer (this will be specified later). Let $\mathcal{K}$ be a random variable the values of which are subsets of the set $\mathcal{K}(G)$ of all k -gons of the r -graph G . If $K \in \mathcal{K}(G)$ then

$$\text{Prob}[K \in \mathcal{K}] = \frac{\varepsilon}{n^{k-r}}$$

and these probabilities are independent for different $K \in \mathcal{K}(G)$. First consider the edges of G which are not covered by chosen k -gons (more precisely the edges of $E_I = E_I(G)$, where $I \in [\{1, 2, \dots, k\}]^r$ and $G = (V(G), E(G) - \{e; \exists K \in \mathcal{K} e \in K\})$. For a fixed edge $e \in E_I$ the probability that $e \in E_I(G)$ is

$$p_e = \left(1 - \frac{\varepsilon}{n^{k-r}}\right)^{\sigma_r n^{k-r}(1+o(1))} \sim \exp(-\varepsilon \sigma_r).$$

These probabilities are independent for different $e \in E_I$. Thus the probability that there are exactly s edges in E_I which are not covered by any k -gon $K \in \mathcal{K}$ is

$$\begin{aligned} \sum_{E \in [E_I]^s} \prod_{e \in E} p_e \prod_{e \in E_I - E} (1 - p_e) &= \\ &= \binom{\rho n^r(1+o(1))}{s} (\exp(-\varepsilon \sigma_r))^s (1 - \exp(-\varepsilon \sigma_r))^{\rho n^r(1+o(1))-s} (1+o(1))^{\rho n^r}. \end{aligned}$$

For any $\sigma > 0$ let S_δ be the set of all integers such that $0 \leq s \leq \rho n^r(1+o(1)) = |E_I|$ and $|s - (\exp(-\sigma_r \varepsilon)) \rho n^r| > \delta \rho n^r \exp(-\sigma_r \varepsilon)$, then by (3) we get

$$\begin{aligned} \sum_{s \in S_\delta} \binom{\rho n^r(1+o(1))}{s} (\exp(-\varepsilon \sigma_r))^s (1 - \exp(-\varepsilon \sigma_r))^{\rho n^r(1+o(1))-s} (1+o(1))^{\rho n^r} \\ < n^r \exp(-c_1 n^r) < \exp(-c_2 n^r) \end{aligned}$$

for some $c_1 > 0$, $c_2 > 0$ and n sufficiently large.

Thus we get

$$\left. \begin{aligned} \rho_I(\mathbf{G}) &\sim \rho n^r \exp(-\sigma_r \varepsilon), \\ \text{for every } I \in [\{1, 2, \dots, k\}]^k &\text{ with probability greater than} \\ 1 - \binom{k}{r} \exp(-c_2 n^r) &> 1 - \exp(-c_3 n^r), \quad c_3 > 0 \end{aligned} \right\} \quad (4)$$

Here we used again that n is sufficiently large. Now we prove the following:

CLAIM. For every $\delta > 0$, $k > l \geq r$,

$$\left. \begin{aligned} \text{Prob}[\sigma^L(\mathbf{G}) &\sim \sigma^L(G) \exp(-\varepsilon \sigma_r \left(\binom{k}{r} - \binom{l}{r} \right)), L \text{ complete}, \\ |L| = l &> 1 - \exp(-c'_l n) \end{aligned} \right\} \quad (5)$$

(here c'_l is the positive constant depending on the size of L only).

We shall proceed by induction on $k - |L|$. If $|L| = k - 1$ and L is complete then there is $k' \in \{1, 2, \dots, k\}$ (say $k' = k$) and $t(L) = \sigma_{k-1} n(1+o(1))$ vertices $v_1, v_2, \dots, v_{t(L)} \in V_{k'} = V_k$ such that

$$\{v_i\} \cup R \in E$$

for every $i \in \{1, 2, \dots, t(L)\}$ and $R \in [L]^{r-1}$. Let A_L be the event that $[L]^r \subset \bigcup \{E_I, I \in [\{1, 2, \dots, k\}]^k\} = \mathbf{E}$. For a vertex v_i , $1 \leq i \leq t(L)$ denote by B_i the event that all edges $\{v_i\} \cup R$, $R \in [L]^{r-1}$ remain in $\mathbf{E}$. It follows from (α) of Lemma 1 that for every i , $1 \leq i \leq t(L)$ the number of complete k -sets L' containing v_i and moreover such that $[L']^r \cap [L \cup \{v_i\}]^r \neq \emptyset$ and $[L']^r \cap [L]^r = \emptyset$ is bounded from above by

$$\binom{k-1}{r-1} \sigma_r n^{k-r}(1+o(1))$$

and from below by

$$\binom{k-1}{r-1} \sigma_r n^{k-r}(1+o(1)) - \binom{k-1}{r} \sigma_{r+1} n^{k-r-1}(1+o(1)) \sim \binom{k-1}{r-1} \sigma_r n^{k-r}.$$

As deletion of one of such k -gons causes that B_i fails to be true provided A_L holds, we have

$$\text{Prob}(B_i|A_L) \sim \left(1 - \frac{\varepsilon}{n^{k-r}}\right)^{\sigma^L n^{k-r} \binom{k-1}{r-1}} \sim \exp\left(-\binom{k-1}{r-1} \sigma^L \varepsilon\right).$$

Clearly the events $(B_i|A_L)$ are independent for fixed L and different i , $1 \leq i \leq t(L)$.

Thus similarly as in the preceding case we get, using (3) that

$$\text{Prob}[\sigma^L(G) \sim_{\delta} t(L) \exp\left(-\varepsilon \sigma^L \binom{k-1}{r-1}\right)]; \quad L \text{ complete}, \quad L \subset V, \\ |L| = k-1] > 1 - \exp(-c'_{k-1} n),$$

where c'_{k-1} is a positive constant independent on n . Here we have again used the fact that there are only polynomially many choices for L . As $t(L) = \sigma^L(G)$ we are done.

Suppose now that we proved our claim (4) for all L , $k-1 \geq |L| \geq k-j$ where $1 \leq j \leq k-r-1$. Let L' be a fixed complete set $|L'| = k-j-1$, $1 \leq j \leq k-r-1$. Without loss of generality suppose that $L' \cap V_i \neq \emptyset$ for $i = 1, 2, \dots, k-j-1$. Let v'_i , $1 \leq i \leq t(L')$ be all vertices of V_{k-j} such that $\{v'_i\} \cup R \in E$ for every i , $1 \leq i \leq t(L')$ and $R \in [L']^{r-1}$. Let again $A_{L'}$ be the event that $[L']^r \subset E$ and for i , $1 \leq i \leq t(L')$ let B'_i be the event that all edges $\{v'_i\} \cup R$, $R \in [L']^{r-1}$ remain in E . Then, similarly to the preceding case we have

$$\text{Prob}(B'_i|A_{L'}) \sim \exp\left(-\varepsilon \sigma^{L'} \binom{k-j-1}{r-1}\right).$$

The events $(B'_i|A_{L'})$ are independent for fixed L' and different i , $1 \leq i \leq t(L')$. Thus again if $\delta > 0$, we have again by (3) that

$$|\{i, \{v'_i\} \cup R \in E \text{ for every } R \in [L']^{r-1}\}| \sim_{\delta} t(L') \exp\left(-\varepsilon \sigma^{L'} \binom{k-j-1}{r-1}\right) \quad (6)$$

with the probability bigger than $1 - \exp(-c'_{k-j-1} n)$ for $c'_{k-j-1} > 0$. By the induction assumption, i.e. (4), we have the probability that

$$\sigma^{L^i}(G) \sim_{\delta} \exp\left(-\varepsilon \sigma^L \left(\binom{k}{r} - \binom{k-j}{r}\right)\right) \sigma^{L^i}(G) \quad \text{for every } i, \\ 1 \leq i \leq t(L') \text{ (here } L^i = L' \cup \{v_i\}) \text{ is larger than } 1 - \exp(-c_{k-j} n). \quad (7)$$

As the set of all k -gons containing L' in G is the union of all sets of k -gons containing such L^i for which $L^i \subset G$ holds we get, combining (6) and (7) and using $(1+\delta)^2 < 1+3\delta$ for $\delta < 1$, that

$$\sigma^{L'}(G) = \sum_{L^i \subset G} \sigma^{L^i}(G) \sim_{3\delta} \\ \sim_{3\delta} \exp\left(-\varepsilon \sigma^L \left[\left(\binom{k}{r} - \binom{k-j}{r}\right) + \binom{k-j-1}{r-1}\right]\right) \sum_{1 \leq i \leq t(L')} \sigma^{L^i} \\ \sim \exp\left(-\varepsilon \sigma^L \left[\binom{k}{r} - \binom{k-j-1}{r}\right]\right) \sigma^{L'}(G)$$

for $\delta < 1$ with probability exponentially close to 1. As there are only polynomially many choices for L' we infer that

$$\sigma^{L'}(G) \sim_{3\delta} \sigma^{L'}(G) \exp\left[-\varepsilon \sigma^L \left(\binom{k}{r} - \binom{k-j-1}{r}\right)\right]$$

for every L' holds with probability bigger than $1 - \exp(-c'_{k-j-1} n)$. This proves the Claim.

Now we shall examine intersections of chosen k -gons. For a system $\mathcal{K}$ of k -gons let $c(\mathcal{K})$ denote the number of pairs $K^1, K^2 \in \mathcal{K}$ such that $K^1 \cap K^2 \neq \emptyset$. Then the expectation

$$\begin{aligned} E(c(\mathcal{K})) &= (1 + o(1)) \binom{k}{r} \rho n^r \binom{\sigma_r n^{k-r}}{2} \left(\frac{\varepsilon}{n^{k-r}} \right)^2 \\ &\leq \frac{1 + o(1)}{2} \binom{k}{r} \rho \sigma_r^2 \varepsilon^2 n^r, \end{aligned}$$

and thus

$$\text{Prob} \left(c(\mathcal{K}) \leq \frac{3}{4} (\rho \sigma_r^2 \varepsilon^2 n^r) \binom{k}{r} \geq \frac{1}{3} (1 + o(1)) \right).$$

According to (4) we have that

$$\left| \bigcup_{K \in \mathcal{K}} K \right| \geq \binom{k}{r} \rho n^r (1 + o(1) - (1 + \delta) \exp(-\sigma_r \varepsilon))$$

holds with probability bigger than $1 - \exp(-c_3 n^r)$ for $n \geq n(\delta)$. As (4) holds for any $\delta > 0$ and $n \geq n(\delta)$, we can assume that

$$\delta \leq \frac{\sigma_r \varepsilon}{10} \exp(\sigma_r \varepsilon).$$

Thus we have for n sufficiently large

$$\begin{aligned} \frac{c(\mathcal{K})}{\left| \bigcup_{K \in \mathcal{K}} K \right|} &\leq \frac{\frac{3}{4} \binom{k}{r} \rho \sigma_r^2 \varepsilon^2 n^r}{\binom{k}{r} \rho n^r (1 + o(1) - \exp(-\sigma_r \varepsilon) - \frac{\sigma_r \varepsilon}{10})} \\ &\leq \frac{\frac{3}{4} \sigma_r^2 \varepsilon^2}{o(1) + \sigma_r \varepsilon - \frac{\sigma_r^2 \varepsilon^2}{2} - \frac{\sigma_r \varepsilon}{10}} < 2\sigma_r \varepsilon, \end{aligned} \quad (8)$$

with probability at least $\frac{1}{3}(1 + o(1))$. Combining (4), (5) and (8) we get that there exists $\mathcal{K} \in \mathcal{K}$ satisfying (a), (b) and (c).

LEMMA 2. *Let $G = ((V_i)_{i=1}^k, E)$ be a k -partite r -graph satisfying assumptions of Lemma 1. Then there exists a system $\mathcal{S}$ of k -gons of G such that*

- (a) $\left| \bigcup_{S \in \mathcal{S}} S \right| = |E|(1 - o(1))$
- (b) $|\mathcal{S}| \leq |E| \binom{k}{r}^{-1} (1 + o(1)).$

PROOF. Let $\delta \geq 0$ be a given real. We will show that there exists (provided $n = |V_1| = \dots = |V_k|$ is sufficiently large) a system of k -gons of G which

$$\text{contains all but at most } \delta/2 |E| \text{ edges} \quad (9)$$

and such that

$$|\mathcal{S}| \leq \frac{|E| \left(1 + \frac{\delta}{2}\right)}{\binom{k}{r}}. \quad (10)$$

We shall construct our system $\mathcal{S}$ inductively. We shall also construct an auxiliary sequence $G_0, G_1, \dots, G_t$ of r -graphs such that $E(G_0) \supset E(G_1) \supset \dots \supset E(G_t)$ (t will be specified later). Set $G_0 = G$ and $\varepsilon = (\delta/4\sigma_r)$. According to Lemma 1 there exists a system $\mathcal{K}_0 = \mathcal{K}$ of k -gons of G_0 with the properties of Lemma 1. Suppose that we have constructed $G_j = ((V_i)_{i=1}^k, E_j)$, $E_j \subset E$ and a system $\mathcal{S}_j$ of k -gons, covering edges of $E - E_j$ so that

$$(a) \quad \sigma^L(G_j) \sim \sigma_l \exp \left[-\sigma_r \varepsilon j \left(\binom{k}{r} - \binom{l}{r} \right) \right] n^{k-l} \text{ for every}$$

L complete in G_j , $k > |L| = l \geq k$,

$$(b) \quad \rho_I(G_j) \sim \rho \exp(-\sigma_r \varepsilon j) n^r \text{ for every } I \in [\{1, 2, \dots, k\}]^r$$

$$(c) \quad |\mathcal{S}_j| \leq \left(1 + \frac{\delta}{2}\right) \frac{|E - E_j|}{\binom{k}{r}}.$$

Then, according to Lemma 1, we can select a system $\mathcal{K}_j$ of k -gons from G_j such that if we put $G_{j+1} = G_j$ —edges of all selected k -gons the following holds:

$$(a') \quad \sigma^L(G_{j+1}) \sim \sigma_l n^{k-l} \exp \left[-\varepsilon \sigma_r (j+1) \left(\binom{k}{r} - \binom{l}{r} \right) \right] \text{ for any } L \text{ complete in } G_{j+1}, k >$$

$|L| = l \geq r$,

$$(b') \quad \rho_I(G_{j+1}) \sim \rho n^r \exp(-\varepsilon \sigma_r (j+1)) \text{ for any } I \in [\{1, 2, \dots, k\}]^r.$$

Moreover, we have

$$|\{\{K^1, K^2\}, K^1, K^2 \in \mathcal{K}_j, K^1 \cap K^2 \neq \emptyset\}| \leq 2\sigma_r \varepsilon \left| \bigcup_{K \in \mathcal{K}_j} K \right|.$$

If we choose for each couple $\{K^1, K^2\}$, $K^1, K^2 \in \mathcal{K}_j$, $K^1 \cap K^2 \neq \emptyset$ at least one of its elements and delete all such k -gons from $\mathcal{K}_j$ we get a system of at least

$$|\mathcal{K}_j| - 2\sigma_r \varepsilon \left| \bigcup_{K \in \mathcal{K}_j} K \right|$$

pairwise disjoint k -gons covering at most $|\bigcup_{K \in \mathcal{K}_j} K|$ edges. Thus we get

$$|\mathcal{K}_j| \leq \frac{|\bigcup_{K \in \mathcal{K}_j} K|}{\binom{k}{r}} + 2\sigma_r \varepsilon \left| \bigcup_{K \in \mathcal{K}_j} K \right|.$$

Set $\mathcal{S}_{j+1} = \mathcal{S}_j \cup \mathcal{K}_j$; as $\mathcal{S}_{j+1}$ covers clearly all edges of $E - E_{j+1}$ we get that

$$\begin{aligned} (c') \quad |\mathcal{S}_{j+1}| &= |\mathcal{S}_j| + |\mathcal{K}_j| \\ &\leq \left(1 + \frac{\delta}{2}\right) \frac{|E - E_j|}{\binom{k}{r}} + \frac{1}{\binom{k}{r}} \left(1 + 2\sigma_r \varepsilon \binom{k}{r}\right) \left| \bigcup_{K \in \mathcal{K}_j} K \right| \\ &< \left(1 + \frac{\delta}{2}\right) \frac{|E - E_j|}{\binom{k}{r}} + \frac{1}{\binom{k}{r}} \left(1 + \frac{\delta}{2}\right) |E_j - E_{j+1}| \\ &\leq \left(1 + \frac{\delta}{2}\right) \frac{|E - E_{j+1}|}{\binom{k}{r}}. \end{aligned}$$

Set $t = \lceil 1/\sigma_r \varepsilon \ln 2/\sigma \rceil$ and repeat this procedure t -times. The system $\mathcal{S} = \mathcal{S}_t$ covers all edges of $E - E_t$ and as $|E_t| \leq \exp(-\sigma_r \varepsilon t) |E| \leq (\delta/2) |E|$ we get that (9) holds. Moreover we have

$$|\mathcal{S}_t| \leq \left(1 + \frac{\delta}{2}\right) \frac{|E - E_t|}{\binom{k}{r}} \leq \left(1 + \frac{\delta}{2}\right) \frac{|E|}{\binom{k}{r}}$$

and thus (10) holds as well.

LEMMA 3. *Let $p > k > r$ be given positive integers. Let $A_1, A_2, \dots, A_p$ be pairwise disjoint sets of the same (large) cardinality n . Then there exists a decomposition*

$$[\{A_i\}_{i=1}^p]^r = \bigcup \{E_J, J \in [\{1, 2, \dots, p\}]^k\} \quad (11)$$

such that for every $J, J' \in [\{1, 2, \dots, k\}]^p, J \neq J'$

(α) $E_J \cap E_{J'} \neq \emptyset$,

(β) $E_J \subset [\{A_i\}_{i \in J}]^r$, and the r -graph $H(J)$ defined by $V(H(J)) = \bigcup_{i \in J} A_i$, $E(H(J)) = E_J$ satisfies

(γ) $\rho_I(H(J)) \sim \frac{1}{t} n^r$, for every $I \in [J]^r$,

(δ) $\sigma^L(H(J)) \sim \sigma_I n^{k-l}$, for any L complete

$$\sigma_I = \left(\frac{1}{t}\right)^{\binom{k}{r} - \binom{l}{r}} \quad \text{and} \quad k \geq |L| = l \geq r,$$

where

$$t = \binom{p-r}{k-r}.$$

PROOF. For every $e \in [\{A_i\}_{i=1}^p]^r$ let $X(e)$ be the set of all k -subsets J of $\{1, 2, \dots, p\}$ with the property that $\{i; A_i \cap e \neq \emptyset\} \subset J$. Clearly $|X(e)| = t$. To every $e \in [\{A_i\}_{i=1}^p]^r$ choose independently on all other choices an element $\phi(e)$ of $X(e)$. For every $J \in [\{1, 2, \dots, p\}]^k$ let $H(J)$ be a random r -graph defined by

$$V(H(J)) = \bigcup_{i \in J} A_i$$

$$E(H(J)) = E_J = \{\phi(e) \subset \{A_i\}_{i \in J}; e \in [\{A_i\}_{i \in J}]^r\}$$

Obviously

$$[\{A_i\}_{i=1}^p]^r = \bigcup \{E_J; J \in [\{1, 2, \dots, p\}]^k\} \quad (12)$$

yields a partition satisfying (α) and (β). We show that with probability $1 - o(1)$ the partition (12) satisfies (γ) and (δ) as well. Let $L, |L| = l < k$ be a fixed subset of $\bigcup_{i \in J} A_i$ such that $|L \cap A_i| \leq 1$ for every $i \in J$. Let A_L^J denote the event that L is complete subset of $H(J)$ and for $v \in A_{i_0} (i_0 \in J - \{i; L \cap A_i \neq \emptyset\})$ let $B_{L \cup \{v\}}^J$ denote the event that

$$[L \cup \{v\}]^r - [L]^r \subset H(J).$$

Then

$$\text{Prob}(B_{L \cup \{v\}}^J | A_L^J) = \text{Prob}(B_{L \cup \{v\}}^J) = \left(\frac{1}{t}\right)^{\binom{k-l}{r}},$$

and these probabilities are independent for different $v \in A_{i_0}$ and fixed L and J . Thus, according to (3) we get that for fixed i_0, J, L and $\delta > 0$

$$\left| \{v \in A_{i_0}; [L \cup \{v\}]^r - [L]^r \in H(J)\} \right| - \left(\frac{1}{t}\right)^{\binom{k-l}{r}} n \geq \delta n \quad (13)$$

holds with the probability smaller than $\exp(-c_1 n)$, where $c_1 > 0$ is independent on n . As there are only polynomially many (in n) choices for i_0, J and L (n is large and $p, k, r, \delta > 0$ are fixed) we get that there exists $c_2 > 0$ ($c_2 < c_1$) such that the probability of the event that there is i_0, J and L such that holds is bounded by $\exp(-c_2 n)$. Similarly one can show that there exists $c_3 > 0$ such that the probability that

$$\left| \rho_I(H(J)) - \frac{1}{t} n^r \right| \geq \delta n^r$$

for some I and J is bounded by $\exp(-c_3 n)$. Hence the partition satisfies (γ) and (δ) with probability at least $1 - \exp(-c_3 n) - \exp(-c_2 n)$. Thus for n sufficiently large there exists a decomposition satisfying $(\alpha), (\beta)$ and

$$\begin{aligned} (\gamma') \quad & \left| \rho_I(H(J)) - \frac{1}{t} n^r \right| < \delta n^r, \\ (\delta') \quad & \left| |\{v \in A_{i_0}; [L \cup \{v\}]^r - [L]^r \in H(J)\}| - \left(\frac{1}{t}\right)^{\binom{k}{r}-\binom{l}{r}} \right| < \delta \left(\frac{1}{t}\right)^{\binom{k}{r}-\binom{l}{r}} n, \end{aligned} \quad (14)$$

for any choice of I, J, L and i_0 . As (14) holds for any L , (L is complete in $H(J)$) $|L| = l$, $r \leq l < k$, we get by induction that L is contained in at least

$$(1 - \delta)^{k-l} \left(\frac{1}{t}\right)^{\binom{k}{r}-\binom{l}{r}} n^{k-l} (1 - \delta)^{k-l} \left(\frac{1}{t}\right)^{\sum_{j=l}^{k-1} \binom{j}{r-1}} n^{k-l}$$

and at most

$$(1 + \delta)^{k-l} \left(\frac{1}{t}\right)^{\binom{k}{r}-\binom{l}{r}} n^{k-l}$$

k -gons of $H(J)$. This together with (γ') and with the fact that $\delta > 0$ may be considered arbitrarily small yields (γ) and (δ) .

PROOF OF THE THEOREM. Let $\delta > 0$ be positive real and k, r positive integers. Take

$$p = \left\lceil \frac{2r(r-1)}{\delta} \binom{k}{r} \right\rceil.$$

Take large integer N (without loss of generality we shall suppose that N is divisible by p) and set $n = N/p$. Consider p pairwise disjoint sets $A_1, A_2, \dots, A_p$ of the same cardinality n . Take the decomposition (the existence of which is ensured by Lemma 3)

$$[\{A_i\}_{i=1}^p]^r = \bigcup \{E_J; J \in [\{1, 2, \dots, p\}]^k\}$$

Thus for each $J \in [\{1, 2, \dots, p\}]^k$ we have that the r -graph $H(J) = ((A_j)_{j \in J}, E_J)$ satisfies the assumptions of Lemma 2. Hence we get, that for $N \geq N(\delta, k, r)$ there exists a system $\mathcal{S}(J)$ of k -gons such that

$$(1) \quad \left| \bigcup_{S \in \mathcal{S}(J)} S \right| \geq |E_J| \left(1 - \frac{\delta}{8 \binom{k}{r}} \right)$$

and

$$(2) \quad |\mathcal{S}(J)| \leq |E_J| \left(1 + \frac{\delta}{8} \right) / \binom{k}{r}$$

To every from at most $\left(\delta/8 \binom{k}{r}\right) |E_J|$ uncovered edges e choose a k -subset of $\bigcup_{j \in J} A_j$ which contains e and let ν_J be a set consisting of all k -gons of $\mathcal{S}(J)$ and all chosen k -sets. We have

$$|\nu_J| \leq \frac{|E_J| \left(1 + \frac{\delta}{8}\right)}{\binom{k}{r}} + \frac{|E_J| \frac{\delta}{8}}{\binom{k}{r}} = \frac{|E_J|}{\binom{k}{r}} \left(1 + \frac{\delta}{4}\right).$$

The set $\nu = \bigcup \{\nu_J, J \in [\{1, 2, \dots, p\}]^k\}$ is a system of at most

$$\binom{p}{k} \left(|E_J| / \binom{k}{r} \right) \left(1 + \frac{\delta}{4}\right) \sim \binom{p}{k} \frac{n^r}{t} \left(1 + \frac{\delta}{4}\right) < \left(\binom{pn}{r} / \binom{k}{r} \right) \left(1 + \frac{\delta}{2}\right)$$

k -sets (for the last inequality we used again that n is large) having the property that any $e \in [\{A_i\}_{i=1}^p]^r$ is contained in some element of ν . Now we examine elements of $[A]^r - [\{A_i\}_{i=1}^p]^r$ ($A = \bigcup_{i=1}^p A_i$). There are at most

$$\binom{pn}{r} - \binom{p}{r} n^r < \binom{pn}{r} \left(1 - \left(1 - \frac{r-1}{p}\right)^r\right) < \binom{pn}{r} \frac{r-1}{p} r$$

of them. For each such r -set e choose an k -set (subset of A) that contains e . This gives at most

$$\binom{pn}{r} \frac{r-1}{p} r < \binom{pn}{r} \delta/2 \binom{k}{r}$$

new k -sets. After adding all such k -sets to ν we get an r -dense system $\mathcal{F}$ of k -sets of $\bigcup_{i=1}^p A_i$ such that

$$|\mathcal{F}| < \binom{N}{r} (1 + \delta) / \binom{k}{r}.$$

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